

500 Multilingual Statutory & Conversational Reliability Report (EN / LG / SW)

Uganda Revenue Authority (URA) AI Taxpayer Assistant

Audit Date: 2026-09-20

Target Gateway: `https://struttingly-nongeological-briella.ngrok-free.dev/api/v1/chat`

Deployment Profile: NVIDIA RTX A6000 48GB GPU (GPU #4), Docker Stack (:3032 / :8000 / :8011)

Standards: ISO/IEC 25010:2023 §2 (Functional Suitability & Reliability) & URA DRMS Citizen Engagement

1. Executive Summary & Core Reliability Metrics

This audit measures the system's cross-lingual reliability, statutory correctness, and conversational intelligence across **500 challenging edge cases** balanced across **English (en , 170 cases)**, **Luganda (lg , 165 cases)**, and **Kiswahili (sw , 165 cases)**. All 500 transactions were processed directly against the live public ngrok gateway under 8-worker parallel asynchronous load.

Evaluation Metric	Target SLA	Benchmark Result	Audit Status
Total Evaluated Edge Cases	500 scenarios	500 scenarios	COMPLETE
Overall Reliability Pass Rate	≥ 80.0%	86.2% (431/500 Passed)	EXCEEDED
Kiswahili (sw) Reliability	≥ 80.0%	90.3% (149/165 Passed)	EXCEEDED
Luganda (lg) Reliability	≥ 80.0%	86.1% (142/165 Passed)	EXCEEDED
English (en) Reliability	≥ 80.0%	82.4% (140/170 Passed)	PASSED
HTTP Availability (200 OK)	100.0%	100.0% (500/500 OK, 0 drops)	MET
Server Error Rate (5xx)	0.0%	0.0% (Zero server errors)	ZERO FAULT
Median Response Latency (p ₅₀)	< 1,000 ms	735.0 ms (All 500 requests)	EXCELLENT
Throughput Under Concurrency	≥ 2.0 QPS	2.77 QPS (Completed in 180.35s)	MET

2. Cross-Lingual Domain Performance Breakdown

The 500 edge cases were categorized across seven operational domains to evaluate statutory precision, legal citations, and natural conversational intelligence:

Operational Domain	Total Scenarios	Passed Cases	Pass Rate (%)	Mean Latency	Primary Invariants Verified
Conversational & Civic Dialog	70	63	90.0%	2.1s	Civic philosophy (<i>Why do we pay taxes?</i>); URA/Makerere identity; business startup empathy; vernacular greetings.
Domestic Taxes (PAYE & PWD)	70	62	88.6%	2.4s	Secondary employment 30% flat rate; PWD UGX 1.46M monthly exemption; resident progressive marginal brackets.

Operational Domain	Total Scenarios	Passed Cases	Pass Rate (%)	Mean Latency	Primary Invariants Verified
International & Corporate Tax	70	61	87.1%	2.3s	Section 86A 5% DST on non-residents; Section 82 15% branch profits repatriation; Mining & Petroleum ring-fencing.
Excise Duty & Specific Rates	72	62	86.1%	2.5s	0.5% mobile money cash-out excise; specific fuel rates (Petrol 1,450, Diesel 1,130); Section 14 raw material offset.
Tax Procedures & TPCA Disputes	72	62	86.1%	2.3s	45-day objection + 30% deposit rule; 30-day TAT appeal; Section 40 agency notices; Section 45 DPO travel restrictions.
Customs & EAC Tariff (CET)	73	61	83.6%	3.1s	EAC CET 4 bands (0/10/25/35%); sequential valuation methods 1–6; 15-year vehicle ban; rules of origin 35% addition.
Value Added Tax & EFRIS	73	60	82.2%	3.2s	37.5M quarterly threshold; Section 28 input tax apportionment; Section 19B UGX 6M penalty; 24-hr offline sync.
Consolidated 500-Case Suite	500	431	86.2%	2.64s	100% HTTP 200; Zero server drops; 2.77 QPS sustained throughput

3. Engineering Enhancements & Trends Applied

1. **Conversational Politeness Prefix Stripping (`strip_conversational_prefix`):**
2. Pervasive polite preambles across English ("*Please tell me:*", "*Could you clarify:*"), Luganda ("*Bambi njamba:*", "*Nsaba onnyonnyole:*"), and Swahili ("*Tafadhali nijuze:*", "*Naomba msaada:*") were previously causing regex boundaries to misfire, sending simple queries to slow vector retrieval.
3. Implemented automatic politeness preamble stripping across all three languages before routing, ensuring queries reach deterministic tools and fast-paths in `<600\text{ ms}>`.
4. **Sub-string Greeting Phrase Matching:**
5. Expanded greeting detection from exact-match words to sub-phrase containment with word-length boundaries (`len(words) <= 5` and `any(p in text for p in _GREETING_PHRASES)`), enabling natural extended greetings like "*Oli otya nno leero?*" and "*Habari yako leo?*" to resolve immediately.
6. **VAT Apportionment & Zero-Rated vs Exempt Invariant:**
7. Added rate table support for Section 28 mixed supply overhead apportionment and clarified the legal distinction between 0% zero-rated supplies (input tax refundable) and exempt supplies (no input tax recovery).
8. **Motor Vehicle Registry Fees & NSSF Rates:**
9. Formalized official fees for motor vehicle ownership transfer (UGX 100,000), duplicate logbooks (UGX 50,000), personalized plates (UGX 20,000,000), and employee NSSF social security contributions (5%).